Notes: Bergemann and Välimäki (JPE, 2006)

Dynamic Pricing of New Experience Goods

Contents

1 Big picture

- **Question.** How should a monopolist dynamically price a **new experience good** when consumers are initially uncertain about their own match value and learn only through use?
- **Setting.** A continuous-time monopoly model with a continuum of buyers, independent private valuations, and learning-by-consuming. A purchase yields both **current consumption** and **information** about future desirability.
- **Core challenge.** Current prices do not only govern current sales. They also determine how many uninformed buyers become informed, and therefore shape the future composition of demand. The seller is managing two submarkets at once: an **uninformed** segment and an **informed** segment.
- **Main theoretical result.** All markets can be sorted into one of two cases using a simple comparison: the willingness to pay of an uninformed buyer at the static monopoly price, denoted $\hat{w}$, versus the static monopoly price itself, denoted $\hat{p}$.
  - If $\hat{w} \geq \hat{p}$, the product is a **mass market**. Prices are high initially and then decline over time. The seller essentially **skims** the uninformed buyers early on.
  - If $\hat{w} < \hat{p}$, the product is a **niche market**. Prices are initially low, then jump up to the static monopoly price once a sufficiently large informed clientele has been built. This is a **penetration / introductory-pricing** pattern.
- **Main dynamic intuition.** The good is a bundle of immediate utility and information. A low current price can be used to create future demand by generating information. A high current price can be used to extract surplus from buyers who already value the product highly.
- **Main welfare result.** With purely **idiosyncratic learning**, competitive marginal-cost pricing implements the social optimum. With **social learning**, however, competition may underprovide experimentation because buyers do not internalize the information they create for others; in that case, monopoly may perform better than competition.
- **Why this paper matters.** It gives a clean and tractable theory that rationalizes why some new goods are launched with falling prices, while others use low introductory prices followed by later price increases. It also clarifies when learning creates informational externalities.

2 Data and measurement (key objects)

This is a **theory paper**; there is no empirical dataset. The relevant "objects" are the model primitives, state variables, value functions, and equilibrium statistics.

2.1 Observed economic environment in the model

- **Consumers.** A continuum of ex ante identical buyers. Each has unit demand per instant for a perishable good.
- **Time.** Continuous time, $t \in [0, \infty)$.
- **Seller.** A monopolist with zero marginal cost in the baseline model.

- **Valuation.** Each buyer has an idiosyncratic willingness to pay v, drawn from a continuously differentiable distribution $F(v)$ with support $[v_l, v_h]$.
- **Aggregate uncertainty.** There is none. The distribution F is common knowledge; uncertainty is only about each buyer's own match value.

2.2 Key measured objects inside the model

- **Ex ante mean valuation.**
$$\bar{v} = \int_{v_l}^{v_h} v\, dF(v).$$
This is the average gross value of the product before learning.
- **Static monopoly price.**
$$\hat{p} = \arg\max_{p \geq 0} p[1 - F(p)].$$
This is the benchmark price when all buyers are informed.
- **Static monopoly quantity.**
$$\hat{q} = 1 - F(\hat{p}).$$
- **Learning rate.** Buyers who purchase receive a perfectly informative signal at Poisson rate λ.
- **Discount rate.** Future payoffs are discounted at rate $r > 0$.
- **State variable.** The fraction of informed buyers is denoted by
$$a(t) \in [0, 1].$$
This single state variable summarizes the dynamic composition of demand.

2.3 Dynamic market composition

- If uninformed buyers purchase, the informed share evolves as
$$\dot{a}(t) = \lambda[1 - a(t)].$$
- If uninformed buyers stop purchasing, learning stops and the state freezes:
$$\dot{a}(t) = 0.$$
- Thus the monopolist's core dynamic choice is whether to keep the uninformed in the market long enough to expand the informed customer base.

2.4 Unobservables and timing

- **Privately known after experimentation.** The true v becomes known only after a buyer consumes and a signal arrives.
- **Publicly relevant but not directly observed.** The state $a(t)$ is not directly observable but can be inferred from equilibrium prices and aggregate sales.
- **Timing at each instant.** The seller posts a spot price, buyers decide whether to purchase, some buyers learn, and the composition of the market updates.

2.5 Conceptual applications

- **Pharmaceuticals.** Patients know the population distribution of effectiveness but must try the drug to learn their own match.
- **Consumer goods and services.** The model can also be read as one in which purchase opportunities arrive randomly and buyers learn after first use.
- **Launch strategy.** The paper is fundamentally about the optimal dynamic pricing of newly introduced products.

3 Models and empirical specifications (all equations + interpretation)

3.1 Static benchmark: informed demand and monopoly pricing

Once buyers are informed, they buy if and only if their realized valuation exceeds the current price. The static monopoly benchmark is therefore

$$\hat{p} = \arg\max_{p \geq 0} p[1 - F(p)], \qquad \hat{q} = 1 - F(\hat{p}).$$

- **Interpretation.** This is the textbook monopoly outcome in the informed submarket.
- **Why it matters.** The entire dynamic classification of the market is built around comparing this static monopoly price with the willingness to pay of an uninformed buyer.

3.2 The value of information and market classification

Suppose the seller were to charge the static monopoly price forever. An uninformed buyer then gets current expected utility $\bar{v}$ and a future option value from learning. The willingness to pay of an uninformed buyer at that benchmark is

$$\hat{w} = \bar{v} + \frac{\lambda}{r} \mathbb{E}\big[(v - \hat{p})^+\big],$$

where $(x)^+ = \max\{x, 0\}$.

The paper defines:

- **Niche market:**
$$\hat{w} < \hat{p}.$$
- **Mass market:**
$$\hat{w} \geq \hat{p}.$$
- **Interpretation.** The object $\hat{w}$ summarizes whether the information obtained from experimentation is valuable enough to keep uninformed buyers in the market even when prices are high.
- **Comparative statics.** A higher learning rate λ or a lower discount rate r makes the market more likely to be a **mass market**, because experimentation becomes more attractive.

- **Economic meaning.** Niche versus mass is not about total market size in the literal sense. It is about whether uninformed consumers would buy at the static monopoly benchmark once the option value of learning is taken into account.

3.3 Buyer value functions

Let $p(a)$ denote a Markov pricing rule and let a denote the current fraction of informed buyers.

For an **informed** buyer with realized valuation v, the value function solves

$$r\,V^v(a) = \max\{v - p(a), 0\} + \frac{dV^v}{da}\,\dot{a}. \tag{1}$$

For an **uninformed** buyer, the value function solves

$$r\,V^u(a) = \max\Big\{\bar{v} - p(a) + \lambda\big(\mathbb{E}[V^v(a)] - V^u(a)\big), 0\Big\} + \frac{dV^u}{da}\,\dot{a}. \tag{2}$$

- **Interpretation of (1).** An informed buyer acts myopically given the current price because her own purchase does not affect future aggregate outcomes.
- **Interpretation of (2).** An uninformed buyer values a purchase not only for current expected consumption, $\bar{v} - p(a)$, but also for the informational gain $\lambda(\mathbb{E}[V^v] - V^u)$.
- **Key idea.** The option value of learning is endogenous because it depends on future prices, which are themselves equilibrium objects.

3.4 Seller problem and Markov-perfect equilibrium

If the seller serves both uninformed and informed buyers, her value function is

$$rV(a) = \max_{p(a)\geq 0}\; p(a)\Big[(1-a) + a\big(1 - F(p(a))\big)\Big] + \frac{dV}{da}\,\dot{a}, \tag{3}$$

subject to the incentive constraint that uninformed buyers are willing to purchase:

$$p(a) \leq \bar{v} + \lambda\big(\mathbb{E}[V^v(a)] - V^u(a)\big). \tag{4}$$

If the seller serves **only informed buyers**, then learning stops and the seller solves

$$rV(a) = \max_{p(a)\geq 0}\; p(a)\,a\big(1 - F(p(a))\big). \tag{5}$$

The state law of motion when the uninformed buy is

$$\dot{a} = \lambda(1-a), \tag{6}$$

and when they stop buying,

$$\dot{a} = 0. \tag{7}$$

- **Interpretation.** The seller is balancing two revenue sources: extracting surplus from already informed high-value buyers and creating future informed demand by inducing experimentation today.

- **Equilibrium concept.** The paper solves for a **Markov-perfect equilibrium**, where strategies depend only on the current state a.
- **Why Markov works.** Under the paper's informational structure, the whole dynamic environment can be compressed into the single state variable a.

3.5 Niche market: the stopping problem

In a niche market, $\hat{w} < \hat{p}$. If the seller were to charge the static monopoly price immediately, uninformed buyers would refuse to experiment. So the seller initially offers lower prices in order to build up a base of informed future customers.

Let $\hat{a}$ denote the stopping point, the informed share at which the seller stops serving the uninformed market. The stopping condition can be written as

$$\pi(\hat{p},\hat{a}) - \pi(\hat{w},\hat{a}) = \frac{\lambda(1-\hat{a})}{r}\,\hat{p}\,[1-F(\hat{p})], \tag{8}$$

where $\pi(p,a)$ is the flow profit at price p when the informed share is a.

- **Interpretation of (8).** The left-hand side is the **current profit sacrifice** from charging the low introductory price $\hat{w}$ rather than the static monopoly price $\hat{p}$. The right-hand side is the **future benefit** from creating more informed customers.
- **Proposition 1.** In the niche market,
$$\hat{a} < 1,$$
and $\hat{a}$ is increasing in λ and decreasing in r.
- **Economic meaning.** Faster learning or greater patience makes it worthwhile to keep introductory pricing in place longer.

3.6 Mass market: stopping when the uninformed cease to be marginal

In a mass market, $\hat{w} \geq \hat{p}$. Uninformed buyers are willing to buy even at the static monopoly benchmark, so the seller initially keeps them exactly indifferent and extracts their current surplus.

Before the stopping point, the equilibrium price is pinned down by the indifference condition of the uninformed buyer:

$$p(a) = \bar{v} + \lambda\big(\mathbb{E}[V^v(a)] - V^u(a)\big). \tag{9}$$

Define the flow revenue function while uninformed buyers are still purchasing:

$$\pi(p,a) = (1-a)p + a\big(1-F(p)\big)p, \qquad p \leq p(a). \tag{10}$$

As long as uninformed buyers are the marginal buyers, the seller wants the marginal flow revenue at $p(a)$ to be nonnegative:

$$\frac{\partial \pi(p(a),a)}{\partial p} \geq 0. \tag{11}$$

The stopping point $\hat{a}$ is then characterized by

$$\frac{\partial \pi(p(\hat{a}),\hat{a})}{\partial p} = 0. \tag{12}$$

- **Interpretation.** At $\hat{a}$, the uninformed buyers stop being the marginal customers. After that, they remain in the market as inframarginal buyers, and the seller starts pricing more aggressively to serve the informed segment as well.
- **Proposition 2.** In the mass market,
$$\hat{a} \leq 1,$$
and $\hat{a}$ is decreasing in λ and increasing in r.
- **Economic meaning.** If learning is faster, uninformed buyers become inframarginal sooner, so the seller switches earlier into the mature pricing regime.

3.7 Early-market price dynamics

Before the stopping point, the equilibrium price path satisfies the differential equation

$$\dot{p}(a) = r\big(p(a) - \bar{v}\big) - \lambda\,\mathbb{E}\big[(v - p(a))^+\big]. \tag{13}$$

The corresponding sales before the stopping point are

$$q(a) = (1 - a) + a\big(1 - F(p(a))\big). \tag{14}$$

- **Interpretation of (13).** The current buyer compares the gain from buying today and learning immediately with the benefit of waiting for tomorrow's possibly different price.
- **Proposition 3.** Before the stopping point, $p(a)$ is decreasing in a. Sales are decreasing for small a and convex in a.
- **Economic meaning.** The value of information falls over time, so the premium embedded in the price falls as well.
- **Additional comparative static.** A more spread-out valuation distribution leads to a faster decline in prices because the option value of learning is more important.

3.8 Mature market pricing

Once the stopping point is reached, the price path differs sharply across the two cases.

- **Niche market.** The seller stops serving uninformed buyers. The price jumps up to the static monopoly price and stays there:
$$p(a) = \hat{p} \quad \text{for } a \geq \hat{a}. \tag{15}$$
Because the uninformed no longer buy, learning stops and a remains fixed at $\hat{a}$.
- **Mass market.** The seller continues to serve the uninformed, but no longer keeps them exactly indifferent. The price continues to decline for all $a \geq \hat{a}$ and converges to the static monopoly price:
$$\lim_{a \to 1} p(a) = \hat{p}. \tag{16}$$
- **Proposition 4.** In the niche market, price *jumps up* at the stopping point; in the mass market, price keeps drifting downward and converges to $\hat{p}$.

- **Long-run quantities.** In a mass market, quantities converge to the static monopoly quantity because eventually almost all buyers are informed. In a niche market, quantities remain permanently below the static monopoly level because the seller stops generating new informed buyers once $a = \hat{a}$.

3.9 Social efficiency with idiosyncratic learning

Let c denote marginal cost. The planner's value for an informed buyer is

$$rW^v = \max\{v - c, 0\}, \tag{17}$$

and for an uninformed buyer,

$$rW^u = \max\Big\{\bar{v} + \lambda\big(\mathbb{E}[W^v] - W^u\big) - c, 0\Big\}. \tag{18}$$

- **Interpretation.** The socially efficient rule says: keep selling as long as the current expected surplus plus the information benefit is nonnegative.
- **Key welfare insight.** With purely idiosyncratic learning, there is no informational spillover across buyers. Therefore, a competitive equilibrium with marginal-cost pricing implements the efficient allocation.
- **Monopoly distortion.** Monopoly distorts both **statically** (price above cost) and **dynamically** (because it may stop experimentation too early in niche markets or price too high during the skimming phase in mass markets).

3.10 Social learning

Now suppose buyers are grouped into sets of size k: one member consumes, but any information generated is shared within the group. Then an informed buyer solves

$$rV^v(a) = \frac{1}{k}\max\{v - p(a), 0\} + \frac{dV^v}{da}\,\dot{a}, \tag{19}$$

and an uninformed buyer solves

$$rV^u(a) = \frac{1}{k}\max\Big\{\bar{v} - p(a) + \lambda\big(\mathbb{E}[V^v(a)] - V^u(a)\big), 0\Big\} + \frac{k-1}{k}\lambda\big(\mathbb{E}[V^v(a)] - V^u(a)\big) + \frac{dV^u}{da}\,\dot{a}. \tag{20}$$

- **Interpretation.** The buyer now learns both from her own experimentation and from the experimentation of others in the same group.
- **Competitive inefficiency.** Each individual buyer internalizes only a fraction of the total information generated by experimentation. This creates an informational externality.
- **Proposition 5.** If the expected current value is below cost, $\bar{v} < c$, but experimentation is socially worthwhile, then for sufficiently large k the product is never sold in competition. A monopolist, however, may still launch it because she internalizes some of the future revenue consequences of information diffusion.
- **Big idea.** The earlier welfare ranking can reverse once learning becomes social rather than purely private.

3.11 Extensions: random purchase opportunities, commitment, and turnover

Random purchase opportunities. If consumption opportunities arrive randomly at rate λ, learning on first purchase yields an equivalent problem with transformed parameters. The buyer problems become

$$(r+\lambda)V^v(a) = \lambda \max\{v - p(a), 0\} + \frac{dV^v}{da}\,\dot{a}, \tag{21}$$

$$(r+\lambda)V^u(a) = \lambda \max\left\{\bar{v} - p(a) + \mathbb{E}[V^v(a)] - V^u(a), 0\right\} + \frac{dV^u}{da}\,\dot{a}. \tag{22}$$

This is equivalent to the baseline model with a higher effective discount rate.

Commitment to a price path. With commitment, the niche-market solution changes sharply. The seller can promise low enough future prices to keep all uninformed buyers in the market until they learn. The optimal commitment price in the niche case is constant and solves

$$\bar{w} = \bar{v} + \frac{\lambda}{r}\,\mathbb{E}\big[(v-\bar{w})^+\big]. \tag{23}$$

This raises welfare in the niche market (though the extra surplus accrues to the seller, not consumers). In a mass market, commitment worsens welfare because it allows the seller to sustain even higher prices.

Stationary market with turnover. If new buyers enter and old buyers exit at common rate g, the willingness to pay of new customers becomes

$$\hat{w}_g = \bar{v} + \frac{\lambda}{r+g}\,\mathbb{E}\big[(v-\hat{p})^+\big]. \tag{24}$$

The same mass-market versus niche-market dichotomy survives. In the niche case, the seller may use probabilistically low and high prices in order to maintain a continuing informed clientele.

4 Identification and instruments (what makes the estimates credible)

This is not an empirical identification paper, so there are no instruments in the usual econometric sense. The right question here is: **what assumptions make the results go through, and what features of the model are doing the real work?**

4.1 What fails in naive static intuition

- A static monopoly model misses the fact that a purchase today changes future demand by creating informed buyers.
- A naive static demand estimation using data from both early and mature phases would confound movement along demand with equilibrium movement in the value of information.
- The key object is not just current willingness to pay; it is willingness to pay **including the option value of learning**.

4.2 What structure identifies the niche-vs-mass dichotomy

- The entire dichotomy is pinned down by the comparison between two scalar objects, $\hat{w}$ and $\hat{p}$.
- $\hat{p}$ summarizes the profitability of serving the informed segment only.
- $\hat{w}$ summarizes the private value of experimentation for an uninformed buyer when future pricing is at the static benchmark.
- This is why the model yields a clean and sharp classification without needing a richer state space.

4.3 Why the Poisson-learning assumption matters

- The assumption that a perfectly informative signal arrives at constant Poisson rate λ keeps uninformed buyers **ex ante identical** until they learn.
- This collapses the whole distribution of buyer histories into the single state variable a.
- Without this assumption, one would generally need to track a much richer distribution of posterior beliefs.

4.4 Why the time-consistent equilibrium is tractable

- The paper solves a **Markov-perfect equilibrium** and shows uniqueness in that class.
- The authors also note that the result extends to a larger class of sequential equilibria as long as players condition only on their own histories and aggregate market data.
- This makes the pricing path a genuine equilibrium object rather than an artifact of an arbitrary commitment device.

4.5 Why welfare results depend on idiosyncratic versus social learning

- With idiosyncratic learning, experimentation creates value only for the experimenting buyer, so competitive pricing internalizes everything relevant.
- With social learning, experimentation benefits others too, so competition undersupplies it.
- Monopoly partially internalizes this spillover because the seller captures future revenue from the enlarged informed base.

4.6 Relation to the literature

- Relative to **Shapiro (1983)**, the paper removes the need for myopic consumers and biased expectations and studies a fully dynamic equilibrium with forward-looking buyers.
- Relative to earlier two-period experience-good models, the current framework allows more flexible intertemporal discrimination and generates both penetration and skimming depending on the market environment.
- Relative to later empirical work on experience goods, the paper provides a tractable structural map from learning and pricing primitives to equilibrium price paths.

5 Tables and figures

There are no empirical tables in the paper. The key evidence is theoretical and is organized around the main figures and propositions.

5.1 Figure 1: Stopping point as a function of λ/r

Read:

- The stopping point $\hat{a}$ is not monotone globally in λ/r because the sign flips when the market switches from niche to mass.
- In a **niche market**, higher λ/r raises $\hat{a}$: the seller keeps introductory pricing in place longer because experimentation becomes more valuable.
- In a **mass market**, higher λ/r lowers $\hat{a}$: uninformed buyers become inframarginal sooner, so the seller switches earlier to mature-market pricing.
- At the knife-edge case $\hat{w} = \hat{p}$, the optimal price is constant and $\hat{a} = 1$.

5.2 Figure 2: Equilibrium price in the niche market

Read:

- Prices begin **below** the static monopoly benchmark and drift downward slightly during the introductory phase.
- At the stopping point, price **jumps up** to $\hat{p}$.
- After that jump, the seller serves only the informed segment, so learning stops and the price remains constant.
- This is the clean penetration-pricing picture: sacrifice current margin to build a goodwill clientele, then harvest surplus later.

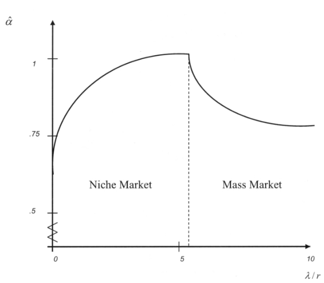

Fig. 1.—Stopping point for varying λ/r

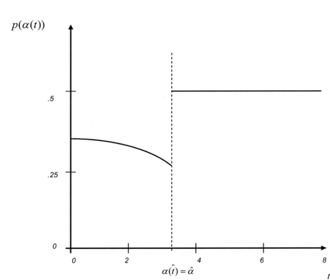

Fig. 2.—Equilibrium price for the niche market

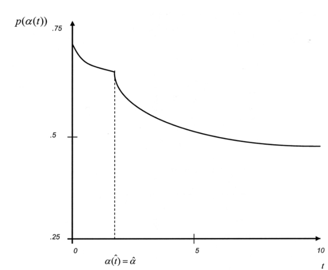

Fig. 3.—Equilibrium price for the mass market

5.3 Figure 3: Equilibrium price in the mass market

Read:

- Prices start high and decline gradually from the very beginning.

- There is no upward jump. Instead, price declines throughout and converges to $\hat{p}$.
- Early on, the seller is effectively skimming uninformed high-value buyers; later, as the informed segment becomes larger, the seller cuts price more aggressively to expand informed demand.
- This is the paper's skimming regime, but generated endogenously from private learning rather than from exogenous optimism.

5.4 Proposition 1 and Proposition 2: stopping rules

Read:

- The two propositions are the backbone of the paper: they characterize how the stopping point behaves in the niche and mass cases.
- The comparative statics go in **opposite directions** across the two regimes.
- This is what gives Figure 1 its hump-shaped logic and is also what makes the market classification economically meaningful rather than a relabeling of the same pricing pattern.

5.5 Proposition 3 and Proposition 4: price-path geometry

Read:

- Proposition 3 says that the early-market price path must decline because the informational option value embedded in the price shrinks over time.
- Proposition 4 then says that what happens at maturity differs completely by regime: a jump up in the niche market, continued decline in the mass market.
- Together, these results provide the full qualitative description of equilibrium pricing.

5.6 Proposition 5 and Section VI: welfare and robustness

Read:

- Proposition 5 shows exactly where the welfare ranking between monopoly and competition can reverse.
- Section VI shows that the central mass-versus-niche dichotomy is robust to buyer turnover and to the alternative interpretation with random purchase opportunities.
- The commitment extension is useful because it clarifies which distortions are due to time inconsistency rather than to monopoly power per se.

6 Short summary

- **Main message.** The optimal pricing of a new experience good depends on whether uninformed buyers are willing to pay the static monopoly price once the option value of learning is included.

- **If the market is a mass market,** prices decline over time and the seller skims the uninformed segment early.
- **If the market is a niche market,** the seller uses low introductory prices to create an informed clientele and then jumps to the static monopoly price.
- **With idiosyncratic learning,** competition is efficient; **with social learning,** monopoly may outperform competition because it internalizes informational spillovers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that dynamic monopoly pricing of experience goods has a simple but powerful structure: every market can be classified as either **mass** or **niche**.
- This classification determines not only whether the seller uses skimming or penetration pricing, but also whether the stopping point reacts positively or negatively to the learning environment.
- The equilibrium is not just a descriptive story about reputation or experimentation. It is a fully dynamic strategic equilibrium with forward-looking buyers and seller.

7.2 Economic intuition

The monopolist is managing information, not just demand. A buyer who experiments today becomes a different kind of customer tomorrow. In a niche market, the seller wants to *create* future customers and is willing to subsidize experimentation through introductory pricing. In a mass market, the seller already has enough willing buyers and instead uses high initial prices to extract surplus from the uninformed. Over time, as more people learn, the relative importance of the uninformed segment shrinks and prices move toward the static monopoly benchmark.

7.3 Takeaway

This paper is best understood as a clean theory of when new products should be launched with *introductory low prices* and when they should be launched with *high prices that then fall over time.* The answer is not an ad hoc marketing rule. It comes from a structural comparison between current monopoly extraction and the future value of creating informed demand. That is what makes the paper so useful: it transforms vague ideas about penetration pricing, skimming, and experience goods into a single equilibrium framework.